

Using advanced machine learning algorithms to predict academic major completion: A cross-sectional study

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ABSTRACT

Background: Existing prediction methods for academic majors based on personality traits have notable gaps, including limited model complexity and generalizability. The current study aimed to utilize advanced Machine Learning (ML) algorithms with smoothing functions to predict academic majors completed based on personality subscales.

Methods: We used reports from 59,413 individuals to perform the current study. All advanced algorithms implemented in this article were based on R software (version 4.1.3, R Core Team, 2021). All model parameters were optimized based on resampling and cross-validation (CV). In addition, pseudo- R^2 as a robust metric has been used to compare the performance of models, which, unlike most studies, considers the quality of model-predicted probabilities.

Result: The results indicated that advanced ML models' performance on training and test data was superior to logistic regression. Pseudo- R^2 and AUC results showed that advanced models such as kNN, GBE, and RF had the highest scores based on test data compared to other models. The pseudo- R^2 values for the models used in this study varied across the test dataset; the lowest value belonged to the logistic regression algorithm at .022, and the highest value was recorded for the kNN algorithm at .099. The agreeableness subscale is the most influential component in predicting the completion of university education, followed by conscientiousness and emotional stability.

Conclusion: The potential of advanced methods to enhance the accuracy and validity of predictions is a promising development in our field. Their performance, particularly in handling large data sets with complex patterns, is a reason for optimism about the future of research in this area.

1. Introduction

One of the most important indicators of education is students' successful completion of academic courses [1]. The quality of the educational system is inherently tied to the successful progression of students through their academic coursework. Enhancing the assessment of educational success through various means, including scrutiny of grade quality, proficiency in the subject matter, and comprehensive fulfillment of course requirements, can elevate the overall quality of the education system [2,3]. In addition to traditional studies that suggest a high

correlation between academic achievement and Intelligence Quotient [1,2,4], contemporary research has demonstrated diverse bio-psycho-social factors influencing academic performance, including levels of scores, Grade Point Average (GPA) level and successful completion of majors [4–6].

Personality is a crucial factor influencing academic achievement. Recent research has shown that certain personality traits can significantly predict academic success [1,7]. Personality encompasses the unique patterns of thoughts, emotions, and behaviors that characterize an individual. It includes how people perceive and interact with the

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world, their emotional responses to different situations, and their consistent behavioral tendencies [8,9]. These patterns are influenced by both genetic and environmental factors, shaping how individuals respond to their environment and interact with others. Furthermore, there is a recent assertion that personality assessments, in isolation, possess substantial predictive power in explaining a significant proportion of the variability observed in academic performance [10,11].

1.1. Machine learning and academic performance

A wide range of studies use different analyses to predict academics according to personality traits/characteristics. For example, one study showed that extraversion and conscientiousness contributed positively to students' academic achievements, primarily through intermediate effects involving self-efficacy and a strong sense of identity in their chosen majors [12]. Another study found that self-efficacy, conscientiousness, openness, and agreeableness significantly and directly contribute to academic performance. Specifically, openness and conscientiousness directly enhance self-efficacy, whereas neuroticism negatively impacts self-efficacy [13]. However, contemporary research increasingly integrates machine learning algorithms to augment predictive accuracy. These approaches address shortcomings of traditional methods, such as regression analysis, which have historically yielded predictions only marginally better than chance [14]. In behavioral science, conventional methodologies like regression have been prevalent for quantifying predictive relationships [15]. Still, the introduction of machine learning offers promising advancements in understanding and predicting complex interactions between personality traits and academic outcomes.

Many advanced models based on machine learning approaches have significantly increased prediction accuracy [16–18]. Machine learning methodologies are progressively utilized in the realm of mental health with dual main objectives: (1) constructing predictive models using supervised learning and (2) assessing and prioritizing variables according to their predictive capability for treatment response [19]. Machine learning has recently experienced considerable progress within behavioral sciences and psychopathology. This approach has a high potential for creating transformation and innovation in existing research methods and clinical interventions, as it can examine large amounts of data and is highly effective in identifying patterns and relationships between variables. The potential value of machine learning algorithms is currently evident and observable, as the computational power of current commercial hardware is compatible with the requirements of machine learning algorithms [20].

Various studies have been conducted in the context of using machine learning algorithms for academic performance. For example, one study used a group of variables such as gender, obesity, academic level, stress level, family size, and lifestyle to predict the GPA, which reflects academic performance [21]. However, this study faced several limitations. Firstly, all participants were residents of a metropolitan city, which may limit the generalizability of the results to rural populations. Additionally, the authors did not evaluate personality disorders or traits, which could be significant factors in academic performance. Similarly, in another study, authors developed a system to identify students at risk of academic difficulties early on, aiming for generalizability across various courses. Their dataset comprised student information from multiple classes and schools, limited to data available at enrollment, excluding post-enrollment performance. The study uniquely included a third intermediate class [22] to tailor different support interventions. They highlighted the limitations of using accuracy as a performance metric for imbalanced datasets, instead employing F1-scores to balance precision and recall for each class. Despite employing various algorithms and addressing class imbalance with techniques like the Synthetic Minority Over-Sampling Technique (SMOTE), the models still struggled with minority class classification. Limitations include the study's focus on a metropolitan sample, potentially limiting generalizability to rural

areas, and omitting personality disorders or traits as predictive factors. Studies have shown that each clinical group has a unique core pathology that is different from other psychiatric disorders [23]. Since a higher prediction of academic performance can help reduce the risk of future success in humans, it is necessary to conduct a more detailed investigation.

Advanced statistical modeling methods prioritize the accuracy of predictions rather than solely focusing on estimating model coefficients. They emphasize the efficiency of prediction accuracy based on new data and exhibit high flexibility and adaptability in identifying complex patterns. So, In the current study, we examine several statistical methods related to machine learning and artificial intelligence that can assist in the field of psychology, as well as their limitations. In summary, we used a collection of advanced machine-learning methods suitable for prediction in psychology. Using advanced machine learning algorithms, one of these was random forest [24]. RF models demonstrate proficiency in addressing issues related to nonlinearity and missed data. Furthermore, they can assign importance scores to each feature variable during classification. This functionality allows for identifying variables that significantly contribute to the classification process [25–27].

In the current study, we used logistic regression, Generalized Additive Model, Decision Tree, Gradient Boosted Tree Ensemble, random forest, Prediction Rule Ensemble, and K Nearest Neighbors have been used. In these advanced models, focusing on the performance accuracy of the model predictions is a higher priority than checking the model coefficients. This is done to minimize the difference between the predicted and observed values of the model. Additionally, many phenomena in the real world exhibit complex patterns and non-linear relationships that are not easily identifiable or detectable based on traditional methods. Nevertheless, more powerful and flexible models can identify more complicated patterns in the data and respond more accurately to intricate relationships within the data. This ability helps mitigate overfitting and results in less bias in the models. However, the formation of a potential contradiction exists. Suppose the complex patterns identified by the models are at least somewhat noisy. In that case, it is possible that the model may not repeat across different iterations from one sample to another and may not be stable enough. This can lead to an increase in variance in the model.

Consequently, the model may react with high sensitivity to new samples. Therefore, pairing powerful algorithms with precise and robust methodologies is essential to improve model performance and balance accuracy and stability. This combination helps prevent model overfitting and balances the bias-variance trade-off. In addition, evaluating the accuracy of the model's performance based on new data is crucial, as models tend to overfit training samples excessively. This approach indicates that traditional analyses are generally insufficient, and machine learning techniques should be used to predict future data under new conditions. This pragmatic approach significantly enhances the model's predictive performance.

According to Barar in 2018, a method for data resampling to evaluate the generalization capabilities of predictive models is the technique of cross-validation. Cross-validation is a critical technique in machine learning as it assesses a model's performance on unseen data and helps detect overfitting. This method reduces the risk of overfitting, and the model's performance is evaluated more accurately. This approach divides data into two sets: training and testing. The model is fitted to the training data, and its performance is assessed on the testing data (de Rooij & Weeda, 2020). Cross-validation allows us to compare different machine-learning methods and understand how well they perform in reality. Therefore, advanced machine learning models have advantages over traditional methods, and in the present study, various algorithms have been compared with each other.

1.2. Study aims

In behavioral science, prediction is a primary objective in leveraging

artificial intelligence, particularly machine learning. Within psychological assessment, tools such as personality questionnaires and various tests gauge and assess individuals' psychological attributes. These conventional development and refinement tools are typically laborious, underscoring the appeal of artificial intelligence and machine learning techniques in psychological evaluation. Therefore, the main objective of this research is to predict the completed field of study based on personality traits. In this study, we consider the independent variable as a subscale score of personality traits and the dependent variable as a binary variable indicating people who studied psychology vs. another field. The current study will focus on something other than prediction's thematic and content aspects. In the test development process, establishing evidence for the validity of scores is crucial, as experts often use it to choose among available tests. Therefore, documenting evidence for the test criterion is an essential component of test construction. For this reason, the capabilities of statistical learning methods should be utilized here.

1.3. Current study importance

The present study made notable advancements in both educational psychology and machine learning by employing sophisticated algorithms to predict academic major completion based on personality traits. Previous research had predominantly relied on conventional statistical methods, such as logistic regression, which often struggled to capture the complexity of the relationships between personality traits and academic outcomes. In contrast, our study utilized advanced machine learning techniques, including kNN, RF, and GBE, which allowed for a more detailed exploration of these intricate patterns. These advanced models significantly outperformed traditional methods in terms of both prediction accuracy and flexibility. While logistic regression assumes linear relationships, the machine learning models used in this study were able to detect non-linear interactions and nuanced correlations between personality subscales and academic outcomes. For instance, traits like agreeableness, conscientiousness, and emotional stability were found to be key predictors of academic major completion, which could not have been as effectively identified through traditional approaches.

The novelty of this study was twofold. First, it demonstrated that machine learning models could handle the complex, high-dimensional data that characterize large-scale educational and psychological datasets. Second, it illustrated the potential for these models to improve predictive accuracy in academic settings, making them valuable tools for educational guidance and decision-making. The study's results underscored the importance of employing these cutting-edge methods to gain deeper insights into the predictors of academic success, particularly in capturing the intricate relationships between students' personality traits and their educational outcomes.

Moreover, this research expanded the applicability of machine learning beyond its conventional domains, showing its relevance in the educational and psychological fields. By demonstrating the superiority of these methods over traditional approaches, the study opened new avenues for integrating data science into academic guidance systems. Ultimately, the findings helped lay the groundwork for developing more personalized and data-driven educational interventions, offering an innovative approach to identifying and supporting at-risk students earlier in their academic journey, thereby enhancing student retention and success rates.

2. Methods

2.1. Participants and procedures

The data from 2015 to 2018 were collected through a website (http://openpsychometrics.org/_rawdata/), where respondents provided information on personality traits and demographic characteristics.

According to this site's privacy policy, while substantial data is gathered and stored for scientific endeavors, it's crucial to emphasize that no efforts will be made to trace users, whether within the site or to other online platforms, through cookies or any other means. All data collected is anonymized. Before beginning the test, participants were informed that their responses would be utilized for research purposes. Upon completion, participants were prompted to confirm the accuracy and suitability of their responses for research. Also, responses that did not meet these criteria have been excluded from the datasets.

In total, 59413 participants had completed a battery of questionnaires including basic demographic questions (age, gender, and country), completed university major, and Ten Items Personality Inventory (TIPI). The mean age of the participants was 29.54, with a standard deviation of 7.33. Additionally, regarding gender, 24,954 participants (42.1 %) were male, and 34,459 participants (57.9 %) were female. Regarding the country of residence, 14,853 participants (25 %) were from the USA, 12,476 participants (21 %) were from the UK, 9506 participants (16 %) were from Germany, 8911 participants (15 %) were from India, 7723 participants (13 %) were from Canada, and 5944 participants (10 %) were from Hong Kong.

The research question aims to investigate whether scores on the subscales of the TIPI can be used to predict the completion of a university major. This study selected individuals who had completed at least one university degree. We consider the dependent variable as binary respondents who studied psychology (15.2 %) or another field (84.8 %).

2.2. Measure

Ten Items Personality Inventory (TIPI): The Ten Items Personality Inventory (TIPI) is a comprehensive self-report instrument specifically crafted to evaluate the Big Five personality traits, namely conscientiousness (C), agreeableness (A), extraversion (E), openness (O), and Emotional Stability (ES). The short version was created and validated in the United States in 2003 by Gosling, Rentfrow, and Swann [28]. Each of the ten items comprises two descriptors, and the measurement of each of the five personality factors is derived from the summation of responses to two questions. The scoring for items in the TIPI is conducted on a 7-point Likert scale, spanning from one (indicating complete disagreement) to seven (indicating complete agreement). The psychometric properties of this questionnaire have been validated [29]. The reason for using this measure of personality, in addition to its brevity, is that it has better quantitative validity than other measures [29]. Table 1 reported the roles of variables used in the machine learning algorithms.

2.3. Model fit and evaluation

We utilized traditional and innovative machine learning/artificial intelligence (ML/AI) methods to investigate the association between sub-scale scores of the Ten-Item Personality Inventory (TIPI) and the

Table 1
Definition of variables used in the models.

variables	Definition
Academic Major Completion	Binary variable indicating whether the individual studied psychology (1) or another field (0)
Conscientiousness (C)	A numerical variable representing a TIPI subscale 1
Agreeableness (A)	A numerical variable representing a TIPI subscale 2
Extraversion (E)	A numerical variable representing a TIPI subscale 3
Openness (O)	A numerical variable representing a TIPI subscale 4
Emotional Stability (ES)	A numerical variable representing a TIPI subscale 5
Age	A numerical variable representing the age of the individual
Gender	A categorical variable indicating the gender of the individual (Male/Female)
Country	A categorical variable indicating the country of the individual (USA/UK/Germany/India/Canada/Hong Kong)

dependent variable. Accordingly, we observed that different algorithms differ in performance. Furthermore, this underscores the importance of researchers having a range of choices and alternatives available. It is crucial to employ distinct and independent datasets for model fitting (training) and evaluation (testing), ensuring the credibility and consistency of the findings. The data was randomly split into two categories: 75 % for training observations and 25 % for testing observations. The training sample for this study included 44,559 respondents, of which 15.1 % chose psychology as their major. Our testing sample consisted of 14,854 respondents, of which 15.4 % chose psychology as their major. This split was chosen due to its appropriate balance between the training and testing data size. This division is commonly used as a standard method in many machine learning and artificial intelligence studies and offers benefits such as having sufficient training data and accurate evaluation. With 75 % of the data allocated for training, the model can effectively utilize the available data for learning and accurately identify patterns. Additionally, 25 % of the data is reserved for testing, which is large enough to accurately and reliably evaluate the model's performance and prevent overfitting.

Other split ratios, such as 70-30 or 80-20, also exist, each with its advantages and disadvantages. The advantage of a 70-30 split is that it allows for more precise model evaluation. Still, the drawback is a reduction in learning accuracy due to the decreased amount of training data. Conversely, the 80-20 split improves the model's ability to learn complex patterns but provides less precise model evaluation due to the smaller test data set. Numerous studies in machine learning have shown that a 75-25 split can create a good balance between training and testing data, establishing this ratio as a standard [30].

In this study, multiple criteria have been used to evaluate the performance of machine learning algorithms. According to Ref. [31], focusing solely on one criterion can distort the reader's interpretation of the comparison between different models. Therefore, it is always preferred to report various criteria. In general, researchers use the AUC (Area Under the ROC Curve) to evaluate machine learning models. This metric is useful for balancing sensitivity and specificity at various thresholds since the area under the curve is defined by sensitivity and specificity of 1. Therefore, the higher the AUC value, the better the algorithm's performance. This study also utilized this metric. However, these criteria do not consider the quality of the predicted probabilities. Therefore, in the present research, we use another metric to assess the accuracy of the predictions. Specifically, we used the Brier score to evaluate the accuracy of the fitted models.

In binary outcome prediction methods, a predicted probability must also be provided in addition to the predicted class to assess classification uncertainty [32]. Indeed, the Brier score, denoted as [25], is a metric used to evaluate the accuracy of probabilistic predictions made by a model. It is defined as the mean squared difference between the observed outcomes and the predicted probabilities:

$$BS = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{p}_i)^2.$$

A lower Brier score indicates better predictive performance, with a perfect score of 0 when predicted probabilities match the actual outcomes.

In the current study, Brier scores were calculated for training and testing observations, with (N) representing the sample size of either the training or testing set. A Brier score equal to the variance (y) of the actual data indicates that the model's performance is no better than random chance (with variances of .1292 for the training data and .1286 for the test data). This signifies the model's failure to predict the data effectively. Moreover, in the current study, the relative measure of pseudo- R^2 was used, which indicates the degree of similarity between the model and the actual data. To calculate the value of pseudo- R^2 , we divide the difference between 1 and the Brier score by the variance of y. This measure has values between 0 and 1, where 0 indicates better

performance than random chance, and 1 indicates perfect prediction accuracy. In other words, this measure indicates how better the model has predicted the data than a random model. Additionally, pseudo- R^2 values are lower than R^2 values, such that R^2 values between .5 and .6 are roughly equivalent to pseudo- R^2 values between .10 and .12.

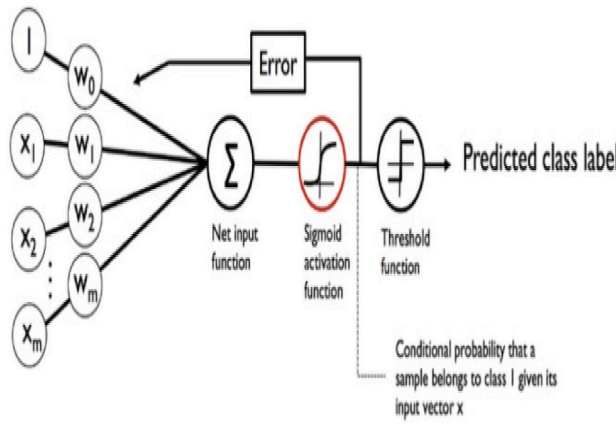
This study also used the z statistic to examine significant differences between the models' performances. This involves calculating the difference in pseudo- R^2 values between two models relative to the standard error of the reference model. If the z statistic value exceeds 1.96, it indicates a significant difference between the two models in prediction.

2.4. The machine learning algorithms in this study

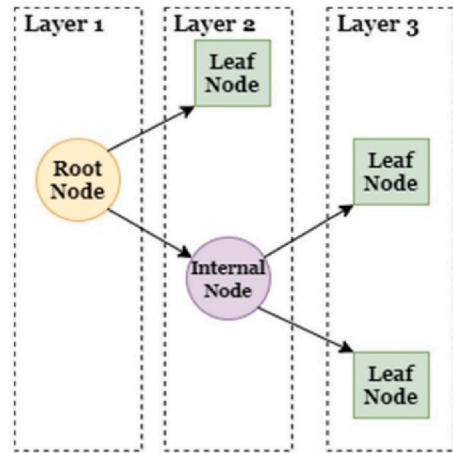
This study uses seven machine learning algorithms to predict the response (dependent) variable. Below, we will examine each of these algorithms, the reasons for their selection, and their suitability for this study. It is worth noting that the methods discussed in this study are known as supervised learning methods.

- **Logistic Regression:** This algorithm, one of the most well-known machine learning algorithms for predicting binary variables, seeks to find a logistic function to model the probability of an event occurring. The simplicity, ability to model linear relationships between variables, and high interpretability are among the advantages of this algorithm and the reasons for its selection in this study.
- **Generalized Additive Models (GAMs):** GAMs are regression models that allow us to model non-linear relationships between input and output variables collectively. This algorithm offers more flexibility in modeling complex relationships between features and the target variable. Given the existence of complex non-linear relationships in the data of this study, GAMs seem to be an appropriate choice.
- **Decision Tree:** This algorithm uses a tree structure for decision-making and can model data complexities simply and interpretably. The ability to model non-linear relationships and interactions between variables is one of the strengths of this algorithm. Decision trees are suitable for data with hierarchical structures and complex feature interactions.
- **Gradient Boosting Trees Ensemble (GBE):** GBE is an ensemble method that helps improve the final model's performance by sequentially combining multiple decision tree models. GBE was chosen due to its high ability to reduce error and increase model accuracy. Given this study's need for high accuracy in this study, GBE seems suitable due to its strong capability to combine models and reduce error.
- **Random Forest (RF):** This algorithm is also an ensemble method that helps improve model accuracy by combining multiple decision trees and creating diversity through bagging. High stability and reduction of overfitting are among the advantages of this algorithm and the reasons for its selection in this study. RF is suitable for noisy data because it prevents overfitting using multiple decision trees.
- **Predictive Rule Ensembles (PRE):** PRE is an algorithm that helps create the final model by combining multiple predictive rules. High interpretability and the ability to model complex relationships are strengths of this algorithm and reasons for its selection in this study. PRE is an appropriate choice for data that require high interpretability.
- **K-Nearest Neighbors (kNN):** This algorithm predicts the target variable based on the distance between samples and their k-nearest neighbors. Simplicity and the ability to model local relationships between data are advantages of this algorithm and reasons for its selection in this study. kNN is suitable for data with a regional structure and requires accurate predictions based on neighborhood.

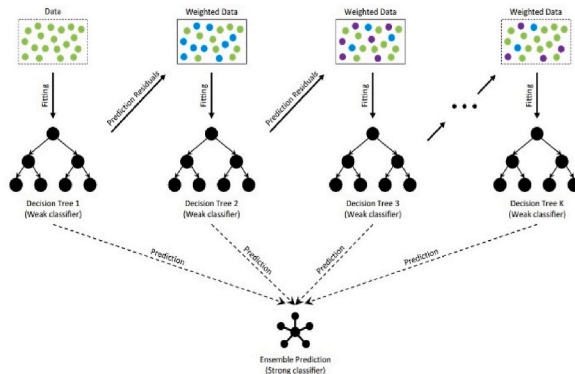
The architecture of the machine learning models in this study is shown in Fig. 1. Given that both logistic regression and GAM models use



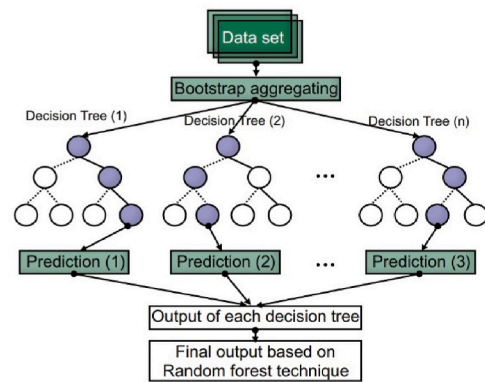
Architecture of Logistic Regression Model [70]



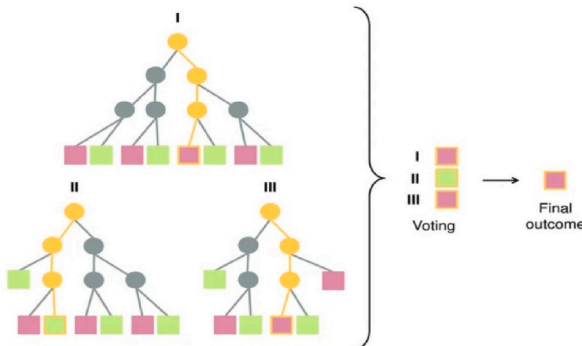
Architecture of Decision Tree Model [71]



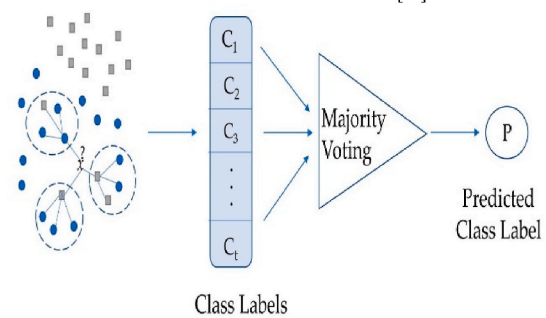
Architecture of Gradient Boosting Trees Ensemble Model [72]



Architecture of Random Forest Model [73]



Architecture of Prediction Rule Ensemble Model



Architecture of k-nearest neighbors Model [74]

Fig. 1. Architecture of machine learning models.

a link function to connect predictions to the dependent variable, the architecture of GAM was not illustrated.

2.5. Comparison of traditional methods and advanced machine learning algorithms

In psychological prediction, various methods are employed to forecast different psychological variables. One standard traditional method is logistic regression, favored for its simplicity and high interpretability in predicting binary variables such as inclination towards specific behaviors or susceptibility to mental disorders. However, logistic regression has its limitations, including.

- Inability to model complex non-linear relationships: Logistic regression assumes a linear relationship between independent and

dependent variables, which is often unrealistic as psychological relationships are frequently complex and non-linear.

- Inability to capture interactions between variables: Logistic regression deals with each independent variable separately from the dependent variable, ignoring interactions between variables. In reality, psychological variables can interact and influence each other.

To transcend the constraints of conventional methods, this study harnesses several advanced machine learning algorithms. These algorithms represent a specialized category of artificial intelligence tailored to address intricate challenges spanning data science, behavioral sciences, and medicine. Unlike traditional machine learning approaches, these algorithms empower robust and adaptable solutions, harnessing extensive datasets to achieve precise and high-fidelity predictions. For instance, generalized additive models exhibit greater flexibility due to their ability to model non-linear relationships compared to logistic

regression. Decision trees can manage complex interactions between variables while maintaining a simple and interpretable structure. Gradient Boosted Tree Ensembles combine multiple decision trees and optimize them gradually, yielding high performance in prediction tasks. Random Forests improve prediction accuracy by combining multiple decision trees, which is particularly effective in handling noisy data. Prediction Rule Ensembles automatically extract multiple prediction rules and demonstrate high capability for identifying complex patterns. kNN predicts based on the proximity of new data points to trained data points, which is suitable for detecting complex and non-linear patterns.

Using machine learning algorithms in psychological prediction offers numerous advantages, including increased prediction accuracy, interpretability, the ability to model large datasets, and the capability to model complex relationships. These algorithms can analyze intricate patterns in data, which traditional methods may overlook, thereby enhancing their utility in predicting psychological variables with greater precision and insight. This approach not only improves predictive performance but also aids in understanding the nuanced interplay between psychological variables, contributing to more practical applications in psychological research and practice.

2.6. Data analysis

All statistical models implemented in this article were based on R software (version 4.1.3, R Core Team, 2021). We fitted logistic regression models using R package *glmnet*; generalized additive models (GAMs) using package *mgcv*; conditional inference trees using package *partykit*; gradient boosted tree ensembles using package *gbm*; random forests using package *ranger*; prediction rule ensembles using package *pre*; k nearest neighbors using package *class*. The model fitting parameters were tuned for all models using resampling and cross-validation (CV) on the training and testing data. The repeated k-fold cross-validation method was used for hyperparameter tuning. In this approach, the data was divided into ten groups ($k = 10$), and the training and validation process was repeated once on each of these groups. The *caret* package was employed to tune the parameters of random forests, boost tree groups, and rule ensemble sets. In this study, instead of using the *caret* package, the researcher manually and meticulously controlled all stages to estimate the parameters of the Conditional Inference Tree and kNN models. The goal was to achieve finer control over the K-Fold Cross-Validation process and parameter tuning in these two models. In general methods using packages like *caret*, many steps are automated, which, while convenient and user-friendly, may not provide sufficient flexibility and control in specific cases. However, in this approach, folds were manually and meticulously created, models were trained for each parameter combination, and every step of the model evaluation process was managed, resulting in a more thorough analysis and greater flexibility in parameter tuning. This manual approach also enabled a more detailed computation and presentation of results, including metrics like Brier Score for each iteration, tree depth, and number of neighbors (K), ultimately contributing to improving model accuracy.

Therefore, for estimating the tree depth parameter in the Tree model and the number of neighbors (K) in the kNN model, there was a need for more precise control over the K-Fold Cross-Validation process and increased flexibility in parameter tuning. For example, manually adjusting tree depths and other model parameters and assessing their impact could help identify optimal settings. Using the manual method allowed for the manual creation and management of folds, storing model predictions for each fold and parameter combination separately, and conducting a more detailed analysis of results, aspects not directly accessible at this level of detail in *caret*. This approach facilitated a more precise calculation and display of evaluation metrics like Brier Score and aided in identifying the most optimal parameter settings.

3. Results

In the current study, all models were examined based on predictor variables of the subscale scores. Moreover, it is possible to investigate whether the results obtained from different models differ from each other or not.

3.1. Machine learning algorithms

3.1.1. Logistic regression (Penalized)

Initially, logistic regression was used as one of the traditional methods of generalized linear models (GLM). In this study, we employed two popular regularization methods, L1 (lasso) and L2 (ridge), to estimate the hyperparameters of the logistic regression model. L1 (lasso) encourages some features to be eliminated with zero weights, whereas L2 (ridge) reduces the weights of all features uniformly but not to zero. Unfortunately, cross-validation results indicated that applying these two regularization penalties did not improve the model's predictive accuracy. In other words, in this study, using L1 (lasso) or L2 (ridge) to control overfitting in the logistic regression model did not lead to improved predictive performance. Regarding the prediction of sub-scale scores, no optimal penalty was identified, and its optimal value was zero. Fig. 2 reports the estimated coefficients for the sub-scale scores; as expected, given the large sample size of this study, sub-scales A, C, and ES were significant at the $p < .001$ level, while sub-scale O was significant at the $P < .01$ level, but sub-scale E did not have a significant effect. Among the significant variables, the most excellent effect was related to the agreeableness facet, with a positive effect, and the smallest effect was related to openness, with a negative effect.

3.1.2. Generalized additive model

We employed GAMs with smoothing splines to model and fit the data. Smoothing splines are very helpful for accurately and flexibly approximating nonlinear relationships between predictors and responses. Splines provide flexible and smooth estimates for observed data points. However, due to the conditional nature of the estimated effects, an additive structure is also applied to the splines. In other words, the impact of each predictor on the response is examined independently of the other predictors, making it easier to interpret the effect of each predictor on the response. The effective degrees of freedom for extraversion, agreeableness, conscientiousness, emotional stability, and openness were 3.64, 7.82, 1.01, 2.13, and 8.63, respectively, as shown in Fig. 3.

Fig. 3 presented the adjusted splines for the sub-scale scores. Similar to the GLM, we observed positive effects on the subscales of agreeableness and conscientiousness and adverse effects on emotional stability and openness. Additionally, the results indicated that all sub-scales exhibited non-linear effects. The spline smoothing within a GAM structure was stable, generalizable, and capable of modeling complex data. Therefore, it enabled inferential analysis methods to predict and accurately model data. In other words, splines helped analyze the data and examined the importance of the effect of each predictor variable in the model. The chi-square test was used in this method to examine the effect of predictor variables. As expected, given the study's large sample size, all sub-scale scores (except for extraversion) had p-values less than .001. In certain datasets, the effect of an independent variable on the dependent variable can vary depending on other predictor variables, making it impossible to examine them separately. In such cases, interactive models are employed to account for interactions between predictor variables when assessing their impact on the response. This approach enables the construction of models that explicitly incorporate interactive effects [see Fig. 2].

3.1.3. Decision tree

This study used a single decision tree based on the conditional inference decision tree Technique [33]. In many decision tree

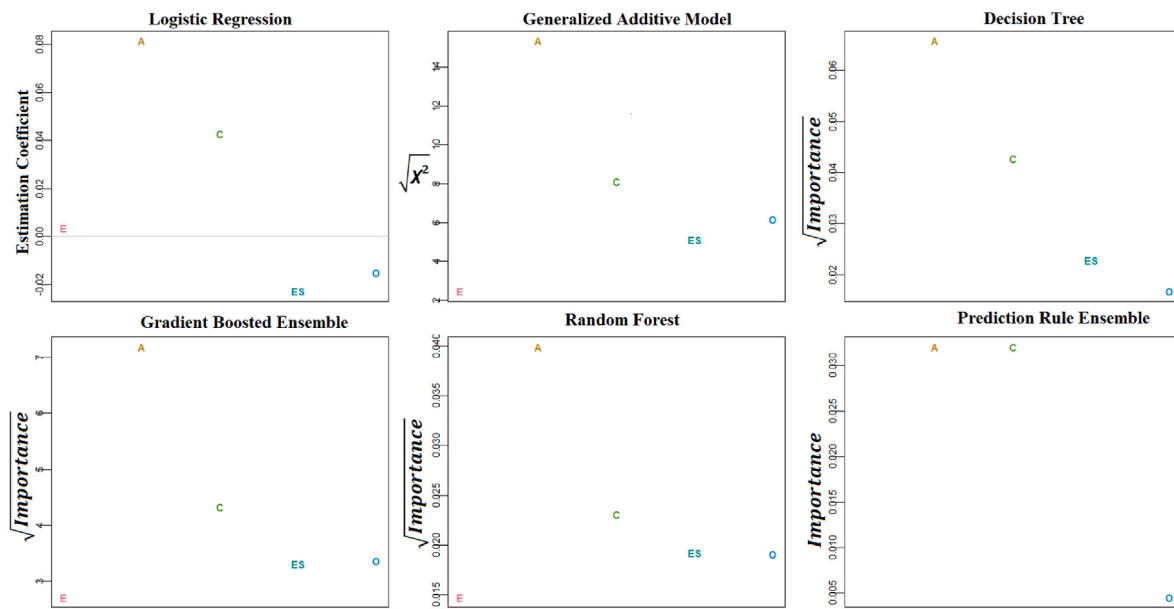


Fig. 2. The importance of predictor variables in each of the fitted models. E = Extraversion; A = Agreeableness; C= Conscientiousness; ES = Emotional Stability; O=Openness.

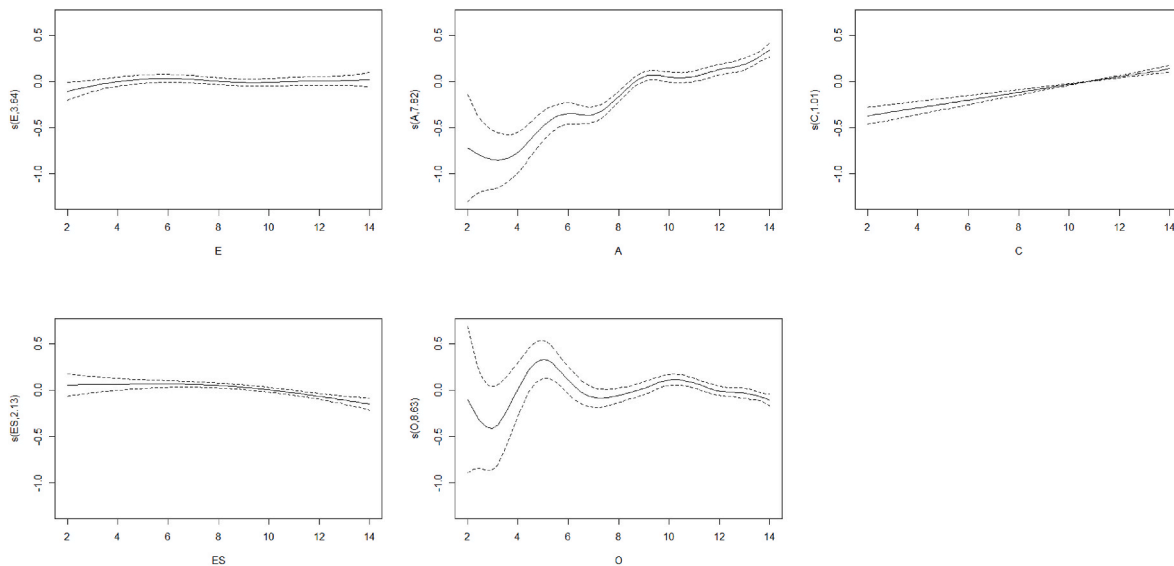


Fig. 3. The adjusted splines for the sub-scale scores. E = Extraversion; A = Agreeableness; C= Conscientiousness; ES = Emotional Stability; O=Openness.

algorithms, there is bias in selecting the following variable for decision-making, leading to suboptimal variable selection. However, this bias is eliminated in this method. In other words, this algorithm selects essential variables in the decision tree using an appropriate and unbiased method, resulting in a better decision tree without biased variable selection [34]. In this study, the optimal parameter maxdepth with a depth of 3 was set for the Decision Tree model to prevent overfitting. However, due to the use of the ctree package for model construction, the parameters minimum number of samples required to split a node and minimum number of samples in each leaf node were automatically optimized by the algorithm itself.

The decision tree constructed based on the sub-scale scores with a depth of 3 is shown in Fig. 4. The sub-scales of agreeableness, conscientiousness, openness, and emotional stability were used in the first split of the tree. The boxes in the terminal nodes indicate the proportion of individuals who have chosen the psychology major in each node.

Therefore, the sub-scales of agreeableness and conscientiousness have a positive effect, while the sub-scale of openness has a negative effect. Regarding Probable interactions, note that divided number 11 indicates that the sub-scale of openness is only relevant for values higher than the sub-scales of agreeableness and conscientiousness. However, such a division may combine collective effects with multiplicity. However, decision trees can incorporate interactive effects; they cannot quickly examine their significance for statistical testing, a disadvantage shared by almost all machine learning and artificial intelligence methods.

Although decision trees are known for their interpretability, they exhibit greater instability than GLMs and GAMs. This sensitivity means their outcomes can vary significantly with minor changes in input data, potentially leading to differing conclusions. Decision trees must generalize effectively to new data, yet their relatively rigid splits contribute to this instability. Combination methods address these issues effectively. In such methods, numerous decision trees (learners) are generated, each

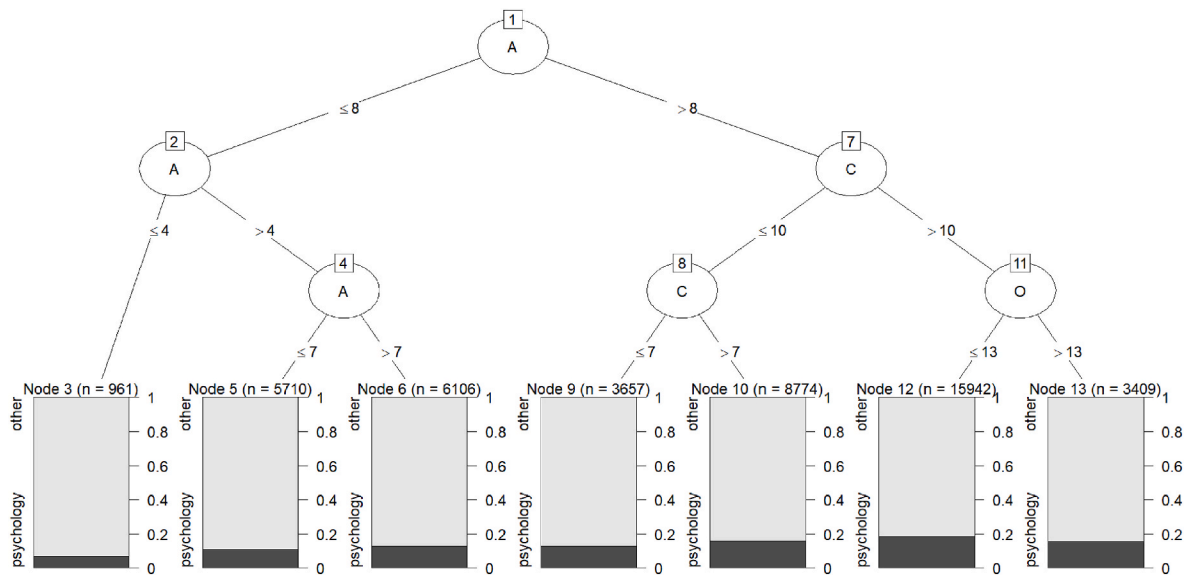


Fig. 4. The decision tree was constructed based on the subscale scores with a depth of 3. A = Agreeableness; C = Conscientiousness; O = Openness.

trained on diverse subsets of training data. The final prediction aggregates the individual trees' predictions, enhancing overall accuracy. These diverse training datasets are created using random techniques like bootstrapping, where data subsets are randomly selected and repeatedly sampled with replacement. This approach, commonly called bagging (see Fig. 2), mitigates decision tree instability and improves predictive performance.

3.1.4. Gradient boosted tree ensemble

In this approach, weak learners are combined sequentially to form strong learners. Weak learners are straightforward models with slightly better prediction accuracy than random ones. In the tree boosting method, weak models are small trees with few splits. In other words, each subsequent tree is adjusted to match the predictions of the previous trees. Therefore, for constructing the next tree, only the part of the data that had a higher prediction error in the last stage is used as input. This process is repeated until a certain number of decision trees are constructed or the prediction error reaches an acceptable level. Thus, the final output is obtained by combining the outputs of all constructed trees. This method is used for complex prediction problems and processing large data sets due to its high accuracy and generalizability for new data.

The main parameters of the GBE algorithm that were optimized include the learning rate (shrinkage), set to .05 to reduce the impact of each tree in the ensemble and prevent overfitting, the number of boosting stages (n.trees) set to 3500 to determine the total number of trees, and the maximum depth of individual trees (interaction. depth) limited to 3 to mitigate model complexity and overfitting to the training data. These parameters were selected based on their performance in improving the model's predictive accuracy using optimization techniques such as multiple iterations and cross-validation.

One of the drawbacks of decision tree ensembles is their inherently non-interpretable nature of how they work and make precise decisions. Therefore, variable importance is used to interpret decision tree ensembles. Variable importance measures the impact of different variables on the model's predictions. In this article, we use permutation importances proposed by Breiman (2001). In this method, the accuracy of the model's predictions is calculated by randomly changing the value of a variable in the test data. Then, the importance of that variable in the model is calculated by comparing the prediction accuracy before and after the change. If changing the variable value significantly reduces the model's prediction accuracy, it means that the variable is significant for

the model's predictions; otherwise, the importance of that variable is lower. The variable importance of the fitted data using the gradient boosting ensemble method is shown in Fig. 2. Importance measures are typically ranked based on the impact of different variables on the predictions of the ensemble model. Using this ranking, we can identify the critical variables for ensemble predictions and use this information to interpret the model and improve prediction accuracy. However, it is essential to exercise caution when interpreting these measures. Simply relying on importance measures cannot determine which predictors have no impact on the prediction outcome, especially in decision tree models. Some predictors may be added to the model that do not impact the prediction outcome but are incorporated into the model due to their correlation with other predictors during the model construction process. In addition, various methods for calculating variable importance may lead to different results, particularly when predictors are correlated. If there are correlated predictors in the model, swapping the values of these predictors can lead to unreal changes in the model, making it difficult to interpret the prediction results.

3.1.5. Random forest

Another popular method of ensemble trees is the random forest [35]. Like boosted trees, random forests use many decision trees and improve performance through ensemble methods. However, random forests do not use sequential learning: in constructing each tree, the prediction results of other trees are not used, and each tree is built independently with the input data. In the original algorithm by Breiman (2001), trees grew to their maximum size. Subsequent studies have shown that if there are correlated predictors in random forests, building large trees may lead to unstable results.

One of the important and distinctive features of RF is the method used to separate variables (predictors). At every node in each tree, a random sample of " m_{try} " independent variables is considered as potential candidates for splitting. The optimal splitting variable and its corresponding value are then selected from this subset of variables. The introduction of randomness in variable selection is crucial, as it prevents each tree from using the same robust set of predictor variables, resulting in a more diverse set of trees. This diversity enhances the performance of the combined trees. In this study, the number of random variables (m_{try}) selected in each section was set to 5 to control the complexity of the model and improve prediction accuracy. Additionally, the minimum node size (min.node.size) was set to 250 to control the growth of trees and prevent overfitting. These parameters were optimized with the aim

of improving the predictive performance of the model on our dataset. Fig. 2 illustrates the significance of the fitted variables using the random forest method.

3.1.6. Prediction rule ensemble

Predictive Rule Ensembles (PRE) [36] were designed to strike a balance between the robust prediction accuracy offered by decision tree ensembles and the interpretability as well as ease of use associated with generalized linear models [37,38]. Decision tree ensembles often have high prediction accuracy, but due to the high complexity of the tree structure and the large number of nodes and conditions, they need higher interpretability. On the other hand, generalized linear models have greater interpretability, but their prediction accuracy may be lower than decision tree ensembles. Therefore, PREs combine the strengths of decision tree ensembles and generalized linear models to balance prediction accuracy and interpretability. In our study, the hyperparameters for the PRE model were optimized using resampling and cross-validation. The optimized parameters included the sample fraction for training (set to .5), the maximum depth of decision trees (set to 3), and the learning rate (set to .05). These parameters determine the structure and performance of the model, and their optimization was carried out to increase accuracy and prevent overfitting.

By using predictive rule ensembles, one can seek a balance between high prediction accuracy and greater interpretability and take advantage of the strengths of both methods. In this approach, a boosted decision tree ensemble is fitted to the training data set, and each node of each tree is considered an independent rule. For example, as shown in Fig. 4, the membership of node 2 in the tree can be encoded using two simple conditions: $A \leq 4$ and $A \leq 8$ &. The membership of node 12 involves multiple conditions: $A > 8$, $C > 10$, and $O \leq 24$. Each of these nodes can be considered a coded rule, such that if the rule conditions hold for the input data, a value of 1 is output, and otherwise, a value of 0 is output for the rule.

The PRE method was applied to a data set that includes rules obtained from decision trees and the main predictor variables. Table 2 shows two of the most essential rules. Since each rule in PRE is considered a logistic regression model, we can use the estimated coefficients to measure the importance of features in predictions. Features with higher coefficients have the most significant impact on predicting the outcome and can be recognized as the essential features in the model. The results are presented in Fig. 2.

3.1.7. K nearest neighbors

The k-Nearest Neighbors (kNN) method is one of the most flexible and important non-parametric methods. In this method, a specific model is not fitted to describe the data, and instead, the training data is directly used to predict the response. The k nearest training data points are found for each new data point, and their response values are averaged. This method provides the highest flexibility among all prediction methods, as in kNN, no function is defined to describe the relationship between input variables and the response variable, and the response is directly predicted from the nearest training data points. This feature makes kNN very useful for processing data where the relationship between input variables and the response variable is complex.

The kNN method has only one adjustable parameter, k. Increasing k results in considering more observations to calculate the average, and thus, a higher mean of response values is used to predict the outcome. In other words, as k increases, the range of influence of neighbors on the

prediction of new observations increases. Therefore, larger values of k reduce variance but increase prediction error. In other words, as k increases, the model becomes simpler and can be the best choice for processing data with a large number of observations. In the kNN method, all variables have equal weight for calculating the distance between data points, meaning that no weight is assigned to each variable. Therefore, this method does not measure the individual effect of variables on the response variable, and we have no information about the contribution of each variable in the prediction. For this reason, no measure of variable contribution is shown for kNN, and we cannot know how effective each variable is in the prediction. In our study, the hyperparameters for the kNN model were optimized using resampling and cross-validation. The optimized parameters included the value of k (the number of neighbors, set to 250), the distance metric (Euclidean distance was used), and the method of weighting neighbors (uniform weighting was used). These parameters determine the structure and performance of the model, and their optimization was carried out to increase accuracy and prevent overfitting.

3.2. Model comparisons

• Variable contributions

Fig. 2 illustrates the fitted models based on the TIPI subscale scores. The logistic regression coefficients also indicate the direction and magnitude of the variable effect. For other models, only the magnitude of the variable's impact on the response variable is shown, and the direction of the effect is not shown. Fig. 2 shows that the impact of variables is similar in all fitted models, and different variables play an essential role in predicting the response variable, i.e., completion of university education. The agreeableness subscale is the most influential component in predicting the completion of university education, followed by conscientiousness and emotional stability. In contrast, the extraversion and openness subscales have the most negligible impact and significance.

The pseudo- R^2 values obtained from the seven proposed models based on the training and testing data are shown in Table 3. The results of Table 3 indicate that the GBE, RF, and kNN models on the training dataset achieved the best model performance in predicting the response variable using subscale scores. These results were entirely predictable, and given that the pseudo- R^2 values for these advanced models were high during the training phase, the results obtained in the testing phase were also predictable. Furthermore, Tree, PRE, and GAM models showed the highest predictive performance in the training data set.

The performance differences among the seven models used in the study are evident in the training dataset. The z statistic was used to examine significant differences between the models' performances. This was done by calculating the difference in pseudo- R^2 values between two models relative to the standard error of the reference model. By considering the logistic regression model as the reference (due to its status as a conventional algorithm), we compare other models to it. If

Table 2
The two most important rules in the prediction rule ensemble.

Description	Coefficient
$A > 8$ & $C > 7$	.114
$A > 8$ & $C > 10$ & $O \leq 13$	.028

Note. A = Agreeableness; C = Conscientiousness; O = Openness.

Table 3

The pseudo- R^2 values are presented based on the training and testing data.

Method	pseudo- R^2 (Test)	SE (Test)	pseudo- R^2 (Train)	SE (Train)	Rank
GLM	.022	.0122	.0411	.0075	7
GAM	.053	.0101	.0935	.0083	5
Tree	.031	.0104	.0988	.0085	6
PRE	.058	.0108	.0981	.0075	4
GBE	.092	.0112	.1823	.0075	2
RF	.086	.0111	.1825	.0081	3
kNN	.099	.0104	.1432	.0077	1

Note. GLM = logistic regression; GAM = generalized additive model; PRE = prediction rule ensemble; GBE = gradient boosted tree ensemble; RF = random forest; kNN = k nearest neighbors.

the z statistic value exceeds 1.96, it indicates a significant difference between the two models in prediction. Based on this, all six advanced algorithms outperformed the logistic regression algorithm in the training dataset. For example, the z statistic value for the kNN algorithm compared to logistic regression was 13.61, which is statistically significant and higher than 1.96 $((.1432-.022) \div .0075)$. Similarly, the advanced algorithms GBE, RF, and Knn outperformed the tree, PRE, and GAM models in predicting the training data.

As the flexibility of machine learning models increases, the performance gap between the training and testing data also widens. In other words, the more flexible the models become, the more they adapt to the training data, and as a result, their performance on the training data improves. However, their performance on the test data decreases. Models that include the main effects of error are less susceptible to these changes. These results are clearly illustrated in Table 3. It can be observed that the variations in pseudo- R^2 values for the advanced models GBE, RF, and kNN are significant between the training and testing phases. Specifically, the pseudo- R^2 values for the advanced RF and GBE models were lower in the testing phase compared to the training phase and showed more fluctuations (Table 2). Therefore, as expected, generally, methods with higher flexibility (RF, GBE, and kNN) are more likely to lead to overfitting, thus reducing test data performance.

Despite the reduced performance of advanced models on test data, they still show significant differences (except for the Tree model) compared to the reference model (logistic regression) based on the z statistic. In the test dataset, all advanced algorithms (except the Tree model) outperformed the logistic regression algorithm. For example, the z statistic value for the kNN algorithm compared to logistic regression was 6.31, which is statistically significant and higher than 1.96 $((.099-.022) \div .0122)$. Among the advanced algorithms, in order of highest predictive performance, were kNN, GBE, RF, PRE, and GAM. The advanced algorithms GBE, RF, and kNN outperformed the tree, PRE, and GAM models in predicting the test data.

The AUC index and ROC curve were also used to evaluate model performance, and the results are reported in Fig. 5. The AUC results were consistent with the pseudo- R^2 index, indicating that the GBE, RF, and kNN models had the highest predictive accuracy with AUCs above .8. Additionally, the Tree model had the lowest AUC index among the

advanced models, with a value less than .60.

4. Discussion

In this study, the predictive performance of advanced machine learning algorithms in psychological assessment was investigated and compared with traditional method. Given the study's aim of providing statistical modeling, a probabilistic approach using pseudo- R^2 was employed to evaluate the algorithms' performance. Typically, in psychological domains, performance evaluation of models often relies on traditional indices such as confusion matrices, which are primarily suitable for decision-making purposes. However, this study focused on examining the models' performance based on the accuracy of predicted probabilities. Additionally, ROC curve analysis was utilized to compare the performance of machine learning algorithms.

In this study, optimal parameters were determined based on a paired approach of advanced algorithms using cross-validation and resampling techniques to enhance prediction accuracy and prevent overfitting. Initially, cross-validation and resampling methods were employed to identify the optimal parameters of the algorithms using training data subsets. This approach ensured that the model was well-adjusted to the diversity of the data and capable of accurately handling new data. The paired method used in this study facilitated precise parameter tuning, reduced overfitting, increased accuracy, and improved model generalization. Consequently, employing algorithm pairing with cross-validation and resampling significantly enhanced the performance of the models.

In this study, all stages were manually designed and executed to achieve more precise control over the cross-validation process and enhance parameter flexibility for Decision Tree and kNN models. This approach enabled the precise computation and display of evaluation metrics such as Brier Score for each iteration, tree depth, and number of neighbors (k), ultimately contributing to improved model accuracy. This method has been able to significantly impact the predictive performance of the advanced KNN machine learning algorithm compared to the traditional logistic regression method. Therefore, the design and development of manually adjusting the parameters plays an essential role in increasing the accuracy and flexibility of the analysis. It is also very effective in identifying the most optimal parameter.

In this study, the results obtained from the training datasets were entirely predictable, as it was expected that advanced algorithms like GBE, RF, and kNN would perform well in predictions. The pseudo- R^2 values for the models used in this study ranged from .0411 to .1825 on the training datasets, with the lowest value belonging to the logistic regression algorithm and the highest to RF. Additionally, in advanced models like GBE and RF, the changes in pseudo- R^2 values between the training and test phases were greater compared to the logistic regression method, aligning with studies such as Fokkema, Iliescu, Greiff, & Ziegler, 2022. This result was entirely predictable because, with increased model flexibility, the models tend to overfit the training data, leading to improved performance on the training data but decreased performance on the test data. Furthermore, with increased flexibility, the risk of overfitting also increases. In this study, by using cross-validation and resampling techniques, we significantly controlled the risk of overfitting for logistic regression, GAM, GBE, RF, and PRE models automatically using the caret package and for Decision Tree and kNN models manually designed.

In this study, the kNN model achieved the highest prediction performance on the test dataset with a pseudo- R^2 value of .099. This indicates that the detailed manual approach to controlling the k-fold cross-validation process for optimally setting the k parameter was highly effective. This method allowed precise control of the folds and enabled us to store the model's predictions for various values of k. Besides kNN, the GBE and RF models also showed high prediction performance on the test dataset. The z-statistic was used to compare the performance of the models. The results indicate that advanced algorithms like kNN, GBE,

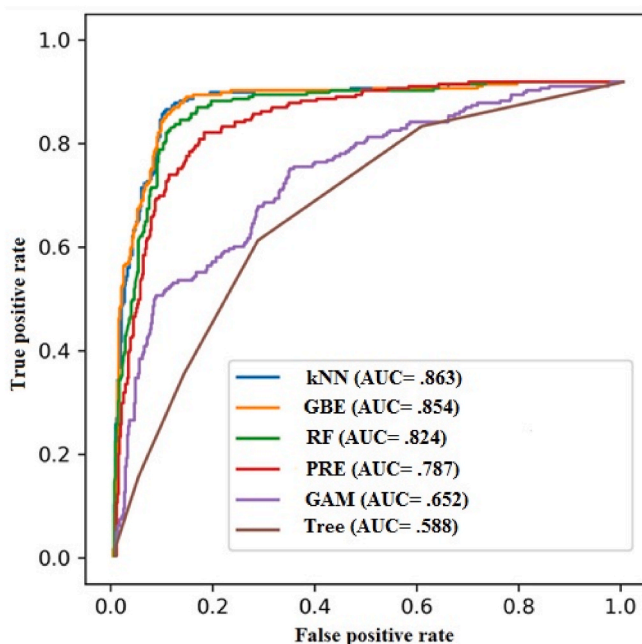


Fig. 5. ROC curve.

and RF performed better than logistic regression on both the training and test datasets. The primary reasons for these performance differences lie in the ability of advanced algorithms to model complex and non-linear interactions between features. While logistic regression fundamentally relies on linear relationships, advanced algorithms such as kNN, GBE, and RF can better identify and model complex non-linear interactions [39]. Moreover, these algorithms leverage more information; for example, kNN uses all data to find the nearest neighbors and can assign more weight to similar data points. GBE and RF, by combining several simple models, can achieve higher accuracy, whereas logistic regression might overfit when dealing with numerous features. Additionally, the greater flexibility in feature selection is a significant advantage of advanced algorithms, allowing them to choose the best features and create more optimized models. In contrast, logistic regression is sensitive to less important features, which can affect its performance. Therefore, advanced algorithms often excel in accuracy and are widely used in many studies due to their superior performance [40–44].

Although advanced machine learning algorithms have significant advantages in modeling complex and non-linear interactions between features, several studies in the fields of psychology and medicine have shown that they do not always outperform linear methods [45–52]. Despite the fact that advanced methods, by incorporating more nonlinear and complex interactions, can create changes in data predictions compared to linear main effects models, the improvements achieved by these methods are often smaller than expected.

Our findings indicate that sub-scale scores provide lower prediction accuracy, which is consistent with previous studies [53,54]. As expected, the value of the explanation coefficient in the test data was lower than in the Train data. Because machine learning algorithms get used to specific patterns in the training data during training, the test data may have patterns or structures that are not present in the training data set. Therefore, the power of prediction in new samples decreases. The results of the present study, in line with previous studies such as [47], indicate the improvement of model predictions based on advanced statistical algorithms.

From the perspective of the relationship between the components of conscientiousness, agreeableness, and extraversion in addition to low neuroticism (high emotional stability) with academic achievements, the present study's findings align with the findings of [55–59]. However, these findings contradict the findings of [60], which indicated an overall negative effect of agreeableness on academic progress.

Although it should be noted that individuals exhibit different levels of these Big Five traits, which interact with each other and influence their attitudes, adaptability, and behaviors, these traits do not operate independently of each other in isolation. However, most studies in this field have reported similar positive effects of conscientiousness, high agreeableness, and low neuroticism (high emotional stability) on academic performance. However, there are differences in the ranking of the impact of these components on academic performance, and most studies have considered the role of conscientiousness more critical. Although conscientiousness remained one of the influential components in the present study, agreeableness had a more prominent role. One reason for this discrepancy can be attributed to the different samples examined in previous studies, and some reasons can be attributed to the statistical tests used, which were mainly conducted using structural equation modeling approaches and small sample sizes or focused on specific aspects of academic performance (such as academic motivation and task completion success).

However, why did agreeableness have a more significant impact on academic performance in the present study? In explaining these results, one can argue that in our research, a fundamental component was examined as the criterion variable, which was the graduation status of students, serving as an outcome of academic performance. It does not focus on the contextual and short-term aspects of academic performance. In other words, although certain personality traits may predict

specific types of academic motivation or performance in certain stages of education, when the graduation status component was examined as the outcome and advanced machine learning methods were utilized, agreeableness and conscientiousness had the highest impact. This finding contains methodological considerations for future studies, indicating that we will likely obtain more diverse results when advanced statistical methods and more fundamental components are employed. These results may not be readily attainable in limited studies with small sample sizes, and one reason for the discrepancy between the present study and studies like Wang et al. (2023) lies in these methodological differences. Such methodological differences could mistakenly conclude that agreeableness hurts academic performance, whereas it has the highest impact on the final academic performance.

In explaining why agreeableness is effective in predicting the graduation status of students, in addition to the methodological considerations mentioned, one can argue that individuals with high agreeableness are cooperative and supportive. They perform better in tasks requiring task division, stability, and support, and they better adapt to the environment [61,62]. They are likely to receive less negative feedback from the educational environment. As a result, such individuals are more likely to complete their education, especially considering that the importance of these components in university is more critical than in school. In universities, there is less emphasis on structured tasks, and education is more collaborative and flexible.

In the next stage, the role of conscientiousness has emerged prominently. In highlighting the importance of conscientiousness in improving academic performance, one can argue that conscientiousness is a personality trait defined by goal-setting, effort, and persistence, and both of these components are associated with better academic performance [63]. This component also contributes to better self-regulation, time management, and organizing efforts toward goals [64]. In other words, individuals with these characteristics possess high responsibility and achievement orientation, effectively predicting successful academic performance. The findings of the present study support the high significance of this component.

4.1. Limitations and future directions

Despite modern statistical prediction methods advancements, their interpretation and referral remain challenging, necessitating robust statistical power and large sample sizes [65]. Referral, which involves assigning importance to individual predictions for decision-making, becomes particularly intricate when numerous predictions exhibit small effects within the dataset. Moreover, the complexity of prediction methods, especially those adept at handling non-linear and interactive effects, further complicates interpretation and referral processes. To mitigate these challenges, various tools have been recommended to enhance the understanding and interpretation of predictions generated by black box models, including sensitivity analysis, critical feature analysis, interactive analysis, and multivariable analysis [66,67]. However, accurately describing the outputs of these tools remains a challenge, potentially leading to misinterpretation and misuse among users unfamiliar with their internal workings [36,68,69]. These methodological considerations underscore the importance of rigorous validation techniques and transparent reporting in studies employing advanced predictive models, such as those investigating the impact of personality traits on academic outcomes. Future research should strive to integrate these tools effectively to improve the reliability and applicability of findings in educational psychology and related fields.

4.1.1. Practical implications

Our findings have profound practical implications for educational institutions and student support systems, particularly in the context of personalized academic guidance. By demonstrating the superiority of advanced machine learning models over traditional methods in predicting academic outcomes, this research underscores the

transformative potential of integrating data-driven approaches into educational settings. These advanced models allow for a more precise analysis of how personality traits influence academic performance, which can lead to the creation of personalized educational interventions tailored to individual student profiles.

Educational institutions can leverage these insights to develop more effective and proactive support systems. For example, machine learning algorithms could be embedded into academic monitoring platforms to automatically identify students at risk of underperforming based on key personality traits like conscientiousness, agreeableness, and emotional stability, which our study identified as significant predictors of academic success. A student showing lower levels of these traits could be flagged for personalized interventions designed to address specific challenges. For instance, if a student exhibits lower conscientiousness, the institution could offer workshops on time management and goal-setting, or provide one-on-one mentoring to enhance organizational skills.

Additionally, these models offer the potential to optimize resource allocation within educational institutions. By accurately predicting which students are most likely to struggle, institutions can direct resources like tutoring, counseling, and academic advising to those who need it most, improving overall efficiency. This predictive capability could also allow schools to design preventative measures that support students before they experience academic failure, reducing dropout rates and improving retention.

The practical applications extend beyond academic performance alone. These predictive models could be used to support student well-being by identifying students who may be at risk of experiencing stress, anxiety, or other mental health issues. By detecting early warning signs through personality profiles and academic behaviors, institutions could provide timely mental health resources, further contributing to student success and well-being.

Furthermore, these machine-learning models can facilitate the development of dynamic academic guidance systems. Instead of relying on static evaluations, these systems could continuously adapt and refine predictions based on a student's ongoing academic performance and behavioral data. This dynamic approach would ensure that interventions remain relevant and effective as students progress through their academic journey.

5. Conclusion

The study results indicate that combining machine learning algorithms with cross-validation and resampling techniques can be effective in real performance evaluation of models. In addition, the findings indicated that with manual control of the entire tuning process (for optimizing parameter settings), significant differences were identified between the performance of advanced and traditional models. Pairing methods can enhance model performance when dealing with complex data and various issues. Moreover, it was observed that increasing the flexibility of machine learning models leads to a greater performance gap between training and testing data. In simpler terms, with greater flexibility, models become more aligned with the training data, improving the performance on the training set.

CRedit authorship contribution statement

Alireza Kordbagheri: Writing – review & editing, Writing – original draft. **Mohammadreza Kordbagheri:** Writing – review & editing, Writing – original draft. **Natalie Tayim:** Writing – review & editing, Writing – original draft. **Abdulnaser Fakhrou:** Writing – review & editing, Writing – original draft, Conceptualization. **Mohammadreza Davoudi:** Writing – review & editing, Writing – original draft.

Availability of data and materials

Data are available in website (https://openpsychometrics.org/_rawdata/).

[_rawdata/](https://openpsychometrics.org/_rawdata/)).

Ethics approval and consent to participate

Considering that the data of this study was received from the online site (https://openpsychometrics.org/_rawdata/), there was no need to receive the code of ethics.

Disclosure statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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